APPENDIX

A IMPLEMENTATION DETAILS

The proposer is Claude Fable 5.1, run through the Claude Code CLI with the `Read`, `Glob`, `Grep`, `Write`, `Edit`, and `Bash` tools. It receives only text at the start of each session and loads images from the filesystem when it needs to inspect them. The search runs for five iterations, and the search and all evaluations run on NVIDIA RTX 6000 Ada GPUs. Table 5 lists the models and their identifiers. The reasoning model is a dated API snapshot rather than a moving alias, open-weight models are pinned to fixed revisions in the released code, and the evaluator decodes greedily. The proprietary models are called through their APIs with all references and the instruction, a square output, and otherwise default settings: GPT-Image-1.5 through the image edit endpoint at 1024×1024, Nano Banana Pro through the Gemini API with a 1:1 aspect ratio, and Seedream 4.5 at its native 2048×2048, downsampled to 1024×1024 before evaluation. Code is available at `https://github.com/KuOnoda/AutoRef`.

Table 5: Model versions used in this work. Identifiers are Hugging Face repositories or API model names. Exact revisions are pinned in the released code.

Model	Identifier
Open image models	
OmniGen2 (Wu et al., 2026a)	`OmniGen2/OmniGen2`
DreamOmni2 (Xia et al., 2026)	`xiabs/DreamOmni2`
BAGEL (Deng et al., 2025)	`ByteDance-Seed/BAGEL-7B-MoT`
FLUX.2 [klein] 4B (Black Forest Labs, 2026)	`black-forest-labs/FLUX.2-klein-4B`
FLUX.2 [klein] 9B (Black Forest Labs, 2026)	`black-forest-labs/FLUX.2-klein-9B`
Qwen-Image-Edit-2511 (Wu et al., 2025)	`Qwen/Qwen-Image-Edit-2511`
DyRef (Huang et al., 2026b)	`Weistrass/Qwen-Image-Edit-2511-DyRef`
Proprietary image models	
GPT-Image-1.5 (OpenAI, 2025b)	`gpt-image-1.5`
Nano Banana Pro (Google DeepMind, 2025b)	`gemini-3-pro-image`
Seedream 4.5 (ByteDance Seed, 2025)	`seedream-4-5-251128`
Reasoning models	
GPT-5.5 (OpenAI, 2026)	`gpt-5.5-2026-04-23`
Qwen3-VL-32B-Instruct (Bai et al., 2025)	`Qwen/Qwen3-VL-32B-Instruct`
Evaluator (MultiBanana)	
Qwen3-VL-8B-Instruct (Bai et al., 2025)	`Qwen/Qwen3-VL-8B-Instruct`
Evaluator (OmniContext)	
GPT-4.1	`gpt-4.1`
Proposer	
Claude Fable 5.1 via Claude Code (Anthropic, 2025)	`claude-fable-5-1`

B EXTENDED RELATED WORK

Test-Time Scaling for Multimodal Generation. Test-time scaling (TTS), which improves model capabilities by allocating additional computation at inference time, has its roots in the development of reasoning in large language models (Kojima et al., 2022; Snell et al., 2025). This paradigm has recently been extended to image and video generation, where a growing body of work improves generation quality and alignment with human preferences (Furuta et al., 2024; Onoda et al., 2026) by scaling inference-time computation without updating model parameters (Yeh et al., 2025; Kim et al., 2025; Ma et al., 2025; Singhal et al., 2025; Oshima et al., 2025; Zhang et al., 2026d; He et al., 2026a). Beyond uniformly increasing computation for all inputs, adaptive TTS dynamically allocates computation budgets according to input difficulty or intermediate evaluations, aiming to achieve more efficient search (Wu et al., 2026b; Zhao et al., 2026b; Guimaraes & Perona, 2026; Rawal et al., 2026; Saini et al., 2026; Jung et al., 2026), with applications spanning image editing (Qu et al., 2026) and joint audio-video generation (Jung et al., 2026). From a broader perspective, self-improving agents

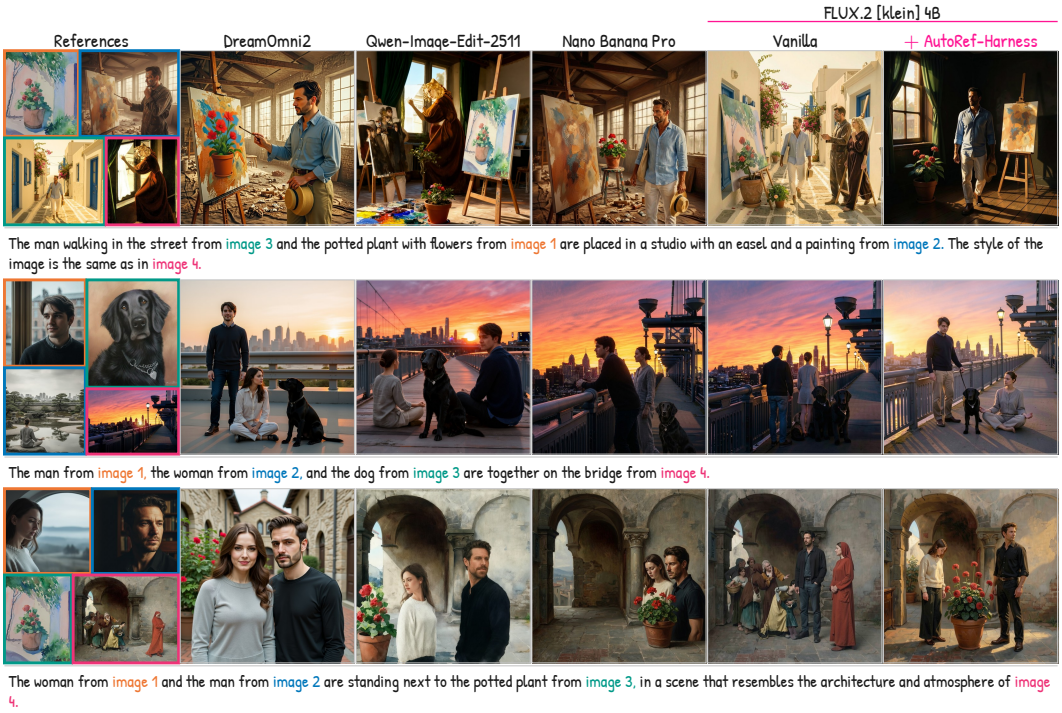


Figure 13: **Background and global style.** Held-out four-reference tasks in which one reference sets the background or the style of the whole image.

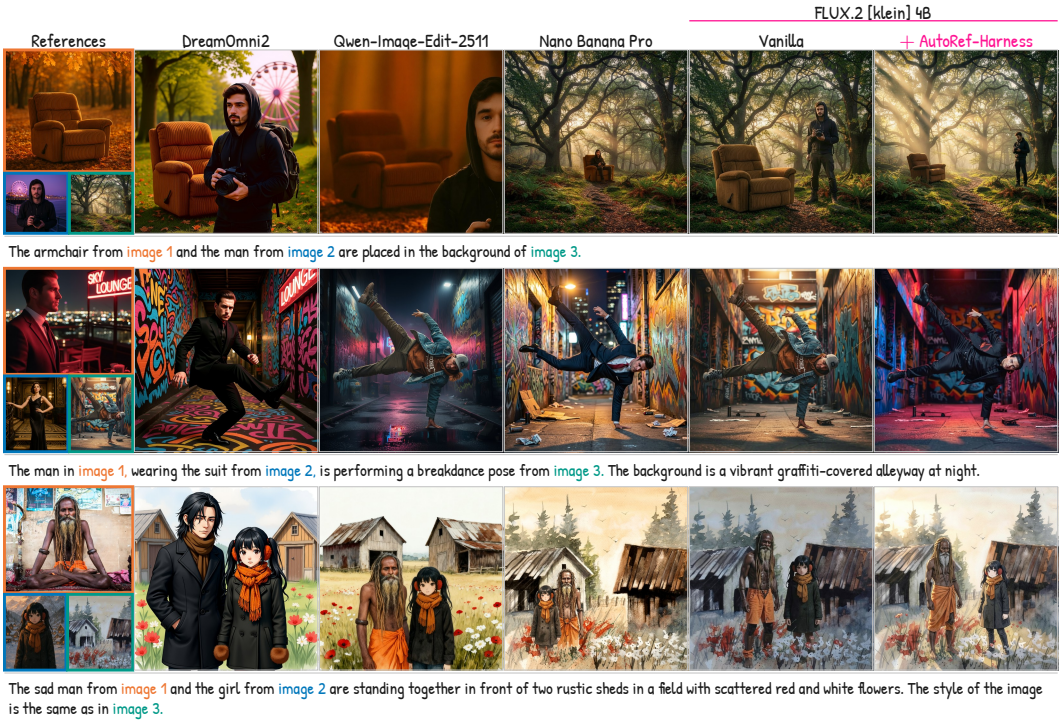


Figure 14: **Three references.** MultiBanana tasks with three references, a count not used during the search.